

Time Series Analysis 2026

Capstone Project Report

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Selected Trading Window: 13 May 2026 – 14 May 2026

Models: LSTM/GRU Hybrid • CNN (Conv1D) • Transformer

Data Source: Yahoo Finance (yfinance) • Framework: TensorFlow / Keras

1. Methodology and Models Used

This project employs three deep learning architectures for daily stock closing price forecasting, all trained on MinMaxScaler-normalized closing prices (Jan 2021 – Dec 2025) from Yahoo Finance using TensorFlow/Keras with a 60-day look-back window:

- **LSTM/GRU Hybrid Model** — A sequential model combining LSTM and GRU recurrent layers. These architectures capture temporal dependencies in time-series data and consistently achieved the lowest error metrics across most stocks in this study.
- **CNN-based Model (Conv1D)** — Uses 1D convolutions with max-pooling and dense layers to extract local short-term patterns from price sequences.
- **Transformer-based Model** — Leverages Multi-Head Self-Attention with Layer Normalization and Global Average Pooling to capture long-range dependencies. This model under-performed relative to the others, likely due to limited training data volume.

2. Stock Selection Rationale

Rather than selecting conventional large-cap stocks (e.g., HDFC, Infosys), a diversified midcap + thematic basket of 7 NSE stocks was chosen for interesting momentum characteristics, sectoral diversity, and non-identical volatility structures to reduce portfolio concentration risk.

Table 1: Selected Stock Universe

Ticker	Sector
DIXON.NS	Electronics Manufacturing
DEEPAKNTR.NS	Specialty Chemicals
ASTRAL.NS	Building Materials
COFORGE.NS	Midcap IT
PERSISTENT.NS	Software Services
ABB.NS	Industrial Automation
APOLLOHOSP.NS	Healthcare

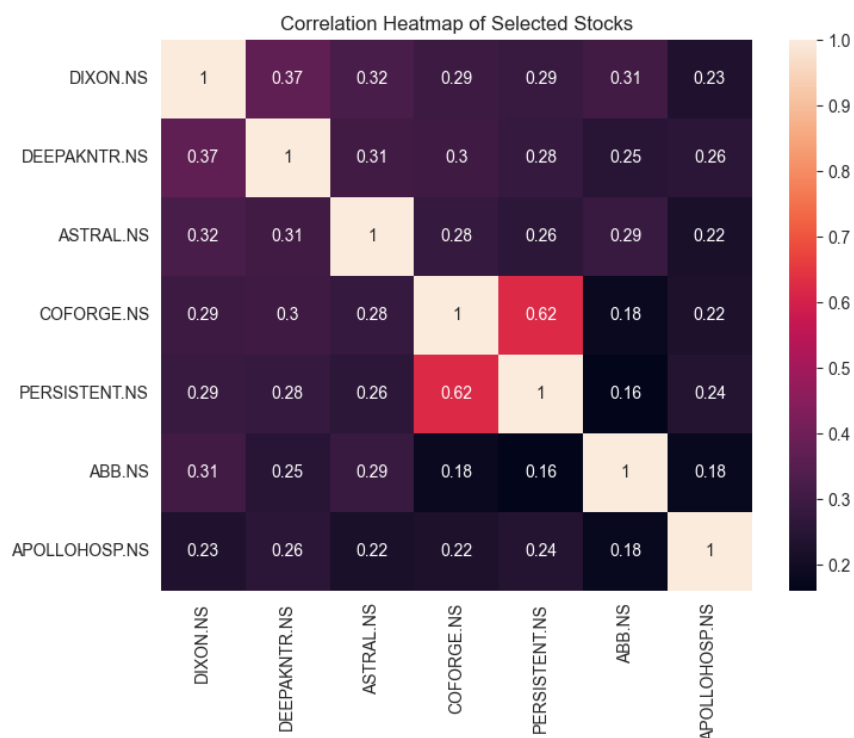
3. Data Preprocessing Steps

- **Data Source:** Daily Close prices from Yahoo Finance (Jan 2021 – Dec 2025).
- **Missing values** handled via forward-fill (ffill).
- **Train/Test Split:** Pre-July 2025 for training; July 2025 onward for testing.
- **Normalization:** MinMaxScaler fitted on training data, applied to both train and test sets.
- **Sequence Creation:** 60-day look-back windows generate input sequences for all three models.
- **Stationary:** ADF tests confirmed non-stationary for most stocks (expected for raw prices). Only DEEPAKNTR showed near-stationary behavior ($p < 0.05$).

Table 2: ADF Stationarity Test p-values

Stock	ADF_p_value
DIXON.NS	0.6861
DEEPAKNTR.NS	0.0300
ASTRAL.NS	0.1269
COFORGE.NS	0.6974
PERSISTENT.NS	0.8776
ABB.NS	0.6009
APOLLOHOSP.NS	0.4320

Figure 1: Correlation Heatmap of Selected Stocks



4. Forecast Results

5-day ahead forecasts were generated for each stock using all three models. LSTM/GRU forecasts served as the primary signal for portfolio decisions. The table shows predicted closing prices (₹) for Days 1–5.

Table 3: 5-Day Forecast Results (₹)

Stock	Model	D1	D2	D3	D4	D5
ABB.NS	CNN	5088.38	5071.85	5056.27	5037.66	5025.65
ABB.NS	LSTM_GRU	5137.48	5130.70	5125.18	5119.80	5114.46
ABB.NS	Transformer	5016.57	5015.46	5014.26	5011.43	5009.89
APOLLOHOSP.NS	CNN	6993.44	6973.10	6959.17	6930.05	6920.54
APOLLOHOSP.NS	LSTM_GRU	7017.65	7002.56	6988.25	6972.94	6956.71
APOLLOHOSP.NS	Transformer	7020.36	7013.90	7006.10	6998.73	6991.76
ASTRAL.NS	CNN	1388.18	1386.07	1384.07	1380.62	1375.91
ASTRAL.NS	LSTM_GRU	1388.23	1385.30	1384.57	1384.53	1384.86
ASTRAL.NS	Transformer	1411.32	1410.69	1410.08	1409.23	1408.52
COFORGE.NS	CNN	1714.90	1696.22	1685.05	1675.61	1657.95
COFORGE.NS	LSTM_GRU	1681.64	1668.12	1657.74	1648.46	1640.08
COFORGE.NS	Transformer	1807.08	1807.63	1806.73	1806.05	1806.31
DEEPAKNTR.NS	CNN	1736.11	1741.22	1743.90	1743.44	1745.43
DEEPAKNTR.NS	LSTM_GRU	1747.34	1754.13	1759.79	1764.59	1768.67
DEEPAKNTR.NS	Transformer	1623.48	1621.87	1620.14	1618.50	1617.03
DIXON.NS	CNN	12293.62	12225.48	12091.89	12062.40	11991.56
DIXON.NS	LSTM_GRU	12127.89	12028.74	11962.80	11906.73	11857.33
DIXON.NS	Transformer	14220.63	14177.15	14136.40	14100.49	14073.33
PERSISTENT.NS	CNN	6210.96	6177.31	6149.44	6115.63	6086.21
PERSISTENT.NS	LSTM_GRU	6210.81	6184.50	6160.94	6137.16	6113.26
PERSISTENT.NS	Transformer	6147.33	6149.12	6146.36	6142.91	6140.40

Figure 2: ABB.NS — 5-Day Forecast

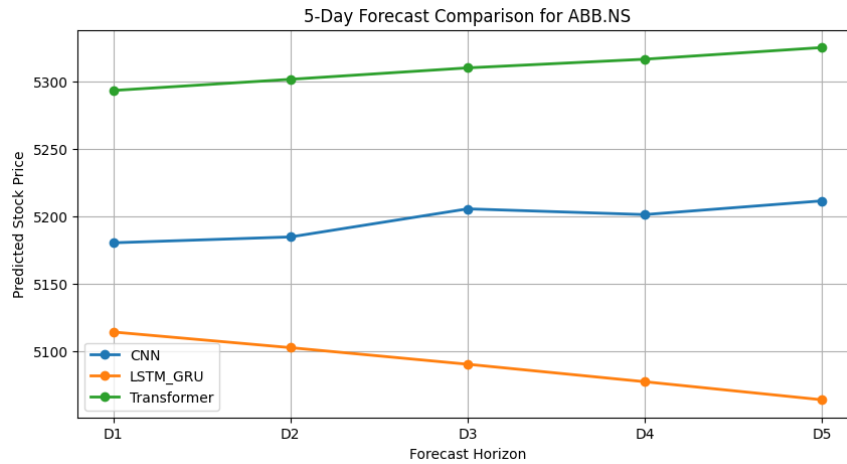
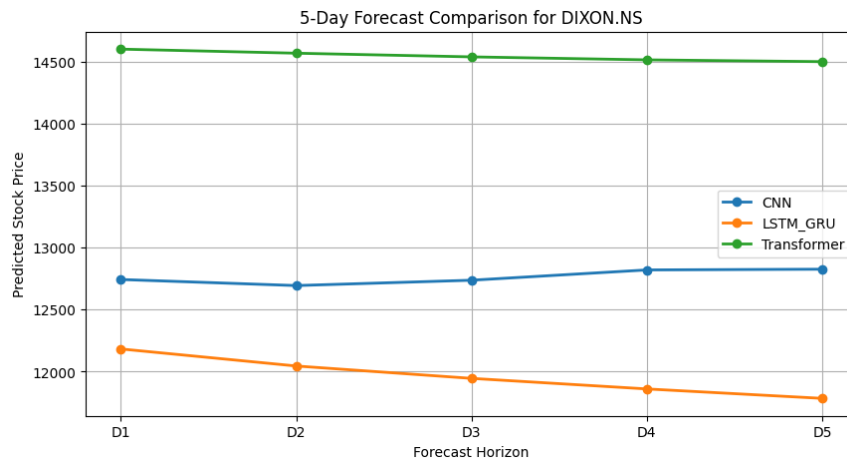


Figure 3: DIXON.NS — 5-Day Forecast



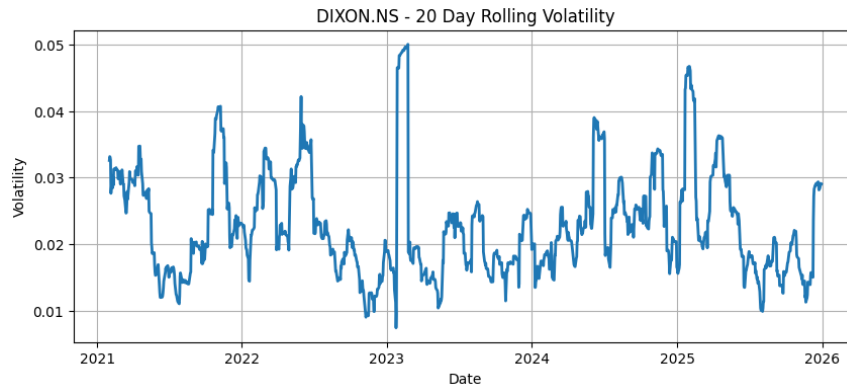
5. Volatility and Trend Analysis

GARCH(1,1) conditional volatility modeling was applied to each stock. STL decomposition (Seasonal-Trend using LOESS) was also performed to separate trend, seasonal, and residual components. DIXON exhibited the highest recent volatility (2.91%), while APOLLOHOSP was the least volatile (0.90%) — directly informing the allocation strategy.

Table 4: GARCH Volatility Summary

Stock	Latest Volatility	Average Volatility	Max Volatility	Min Volatility
DIXON.NS	2.91%	2.27%	5.00%	0.74%
COFORGE.NS	2.03%	2.20%	4.95%	0.86%
PERSISTENT.NS	1.84%	2.11%	3.81%	0.77%
DEEPAKNTR.NS	1.72%	1.99%	4.95%	0.53%
ASTRAL.NS	1.45%	1.85%	4.47%	0.74%
ABB.NS	1.08%	1.94%	4.32%	0.72%
APOLLOHOSP.NS	0.90%	1.65%	3.82%	0.67%

Figure 4: DIXON.NS — GARCH Conditional Volatility



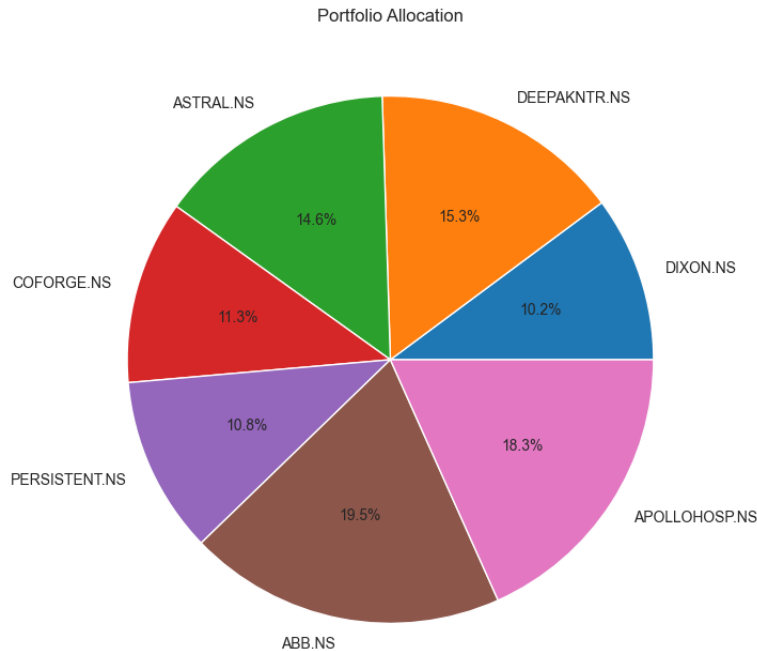
6. Portfolio Composition and Allocation

An inverse-volatility weighting strategy was employed: stocks with lower recent volatility receive larger allocation weights. Total capital: ₹10,00,000. ABB received the largest weight (20.50%) due to lowest recent volatility, while DIXON received the smallest (9.79%).

Table 5: Portfolio Allocation Strategy

Stock	Weight	Allocation
DIXON.NS	9.79%	₹97913
DEEPAKNTR.NS	14.20%	₹142015
ASTRAL.NS	15.21%	₹152108
COFORGE.NS	10.52%	₹105208
PERSISTENT.NS	11.63%	₹116258
ABB.NS	20.50%	₹205010
APOLLOHOSP.NS	18.15%	₹181488

Figure 5: Portfolio Allocation Distribution



7. Model Comparison and Accuracy Evaluation

All three models were compared on the test set using MAPE (Mean Absolute Percentage Error), RMSE (Root Mean Squared Error), and Directional Accuracy (% of correct up/down predictions).

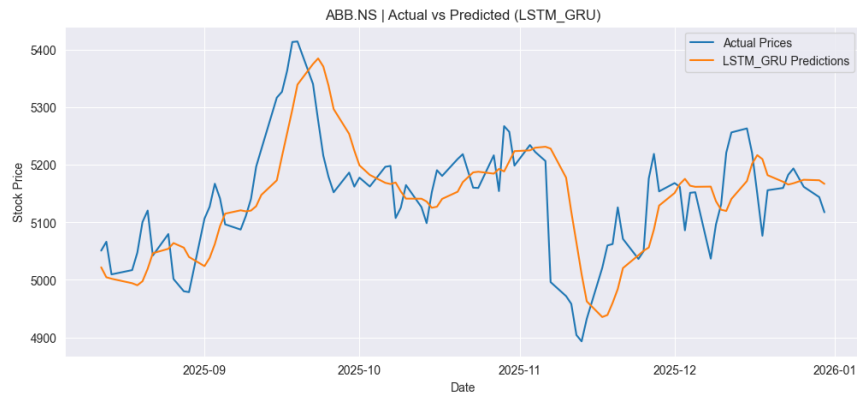
Table 6: Model Accuracy Comparison

Stock	Model	MAPE	RMSE	Directional Accuracy
ABB.NS	LSTM_GRU	1.22%	80.29	47.87%
ABB.NS	CNN	1.46%	95.23	48.94%
ABB.NS	Transformer	9.84%	538.90	43.62%
APOLLOHOSP.NS	LSTM_GRU	1.39%	148.26	58.51%
APOLLOHOSP.NS	CNN	1.57%	162.20	59.57%
APOLLOHOSP.NS	Transformer	4.50%	423.83	55.32%
ASTRAL.NS	CNN	1.65%	30.83	52.13%
ASTRAL.NS	LSTM_GRU	1.80%	34.85	52.13%
ASTRAL.NS	Transformer	3.49%	65.78	40.43%
COFORGE.NS	LSTM_GRU	2.39%	54.07	47.87%
COFORGE.NS	CNN	2.56%	59.31	52.13%
COFORGE.NS	Transformer	5.15%	104.34	45.74%
DEEPAKNTR.NS	CNN	1.77%	36.83	45.74%
DEEPAKNTR.NS	LSTM_GRU	1.85%	37.80	50.00%
DEEPAKNTR.NS	Transformer	4.46%	89.72	47.87%
DIXON.NS	LSTM_GRU	2.57%	481.44	51.06%
DIXON.NS	CNN	3.19%	608.98	52.13%
DIXON.NS	Transformer	8.01%	1558.35	58.51%
PERSISTENT.NS	LSTM_GRU	2.16%	164.71	54.26%
PERSISTENT.NS	CNN	2.47%	190.66	50.00%
PERSISTENT.NS	Transformer	9.42%	631.35	46.81%

Key Observations:

- LSTM/GRU achieved MAPE of 1.2–2.6% — consistently the best performer.
- CNN was comparable but slightly worse (1.5–3.2% MAPE).
- Transformer showed much higher errors (3.5–9.8% MAPE), likely due to insufficient data.
- Directional accuracy was inconsistent across all models, with no clear winner.

Figure 6: ABB.NS — LSTM Test Predictions vs Actual



8. StockGro Execution Summary

Virtual trades were executed on StockGro during the 13–14 May 2026 trading window. Positions were opened at market open on Day 1 based on LSTM/GRU model forecasts, following the inverse-volatility allocation with ₹10,00,000 total capital.

Figure 7: Portfolio — Day 1

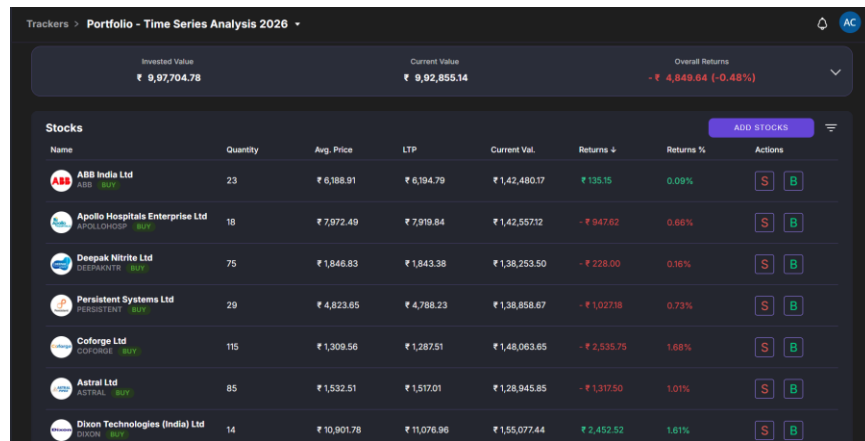
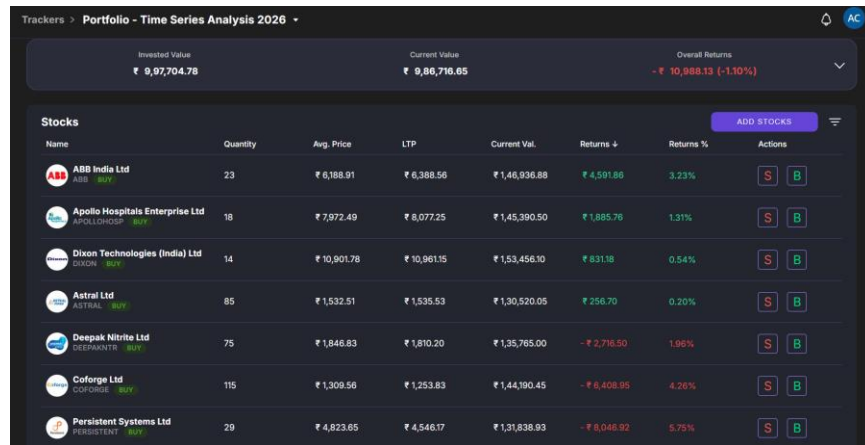


Figure 8: Portfolio — Day 2



9. Predicted vs Actual Outcomes

This is the core evaluation. LSTM/GRU predictions for Days 1–2 of the live trading window were compared against actual closing prices.

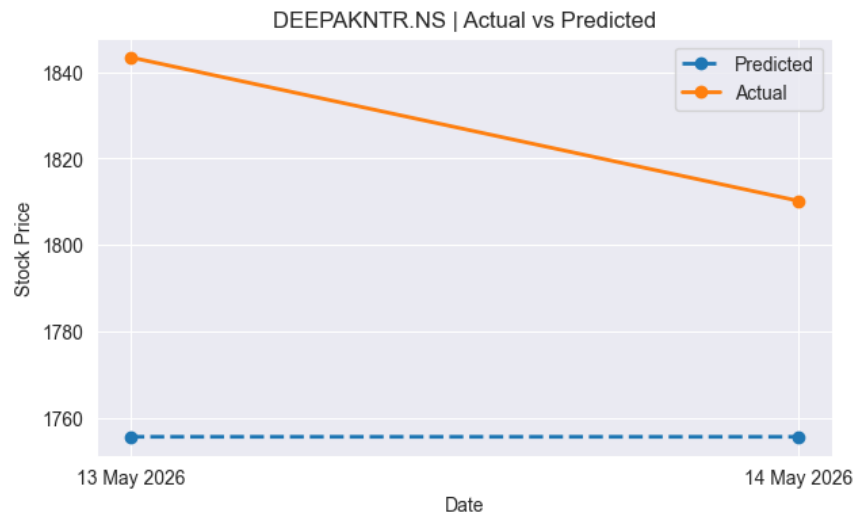
Table 7: Actual vs Predicted — Live Trading Results

Stock	Pred_Day1	Actual_Day1	Pred_Day2	Actual_Day2	Direction_Correct	Live_MAPE
DIXON.NS	12245.55	11076.96	12185.18	10961.15	True	10.86%
DEEPAKNTR.NS	1755.67	1843.38	1755.66	1810.20	True	3.89%
ASTRAL.NS	1364.97	1517.01	1363.70	1535.53	False	10.61%
COFORGE.NS	1680.60	1287.51	1673.37	1253.83	True	32.00%
PERSISTENT.NS	6216.96	4788.23	6193.72	4546.17	True	33.04%
ABB.NS	5154.79	6194.79	5151.39	6388.56	False	18.08%
APOLLOHOSP.NS	7040.49	7919.84	7030.12	8077.25	False	12.03%

Key Findings:

- Direction correctly predicted for 4/7 stocks (DIXON, DEEPAKNTR, COFORGE, PERSISTENT).
- Best live MAPE: DEEPAKNTR at 3.89%. Worst: PERSISTENT at 33.0%, COFORGE at 32.0%.
- ABB, APOLLOHOSP, ASTRAL showed incorrect directional predictions.
- Significant deviation from test-set accuracy highlights real-time prediction challenges.

Figure 9: DEEPAKNTR.NS — Actual vs Predicted (Live)



10. Reflections

What Worked:

- LSTM/GRU hybrid model consistently outperformed both CNN and Transformer on historical test data across all seven stocks, validating its suitability for financial time-series forecasting.
- Inverse-volatility portfolio allocation provided a risk-balanced approach to capital deployment.
- Diverse stock selection across 7 sectors effectively reduced concentration risk.
- Clean preprocessing pipeline (MinMaxScaler, 60-day windows, proper train/test split) ensured robust model training without data leakage.

What Didn't Work:

- The Transformer model underperformed significantly, likely because the self-attention mechanism requires substantially more training data to learn meaningful temporal patterns.
- Live prediction accuracy was considerably worse than test-set accuracy for several stocks (e.g., COFORGE: 2.4% test MAPE vs 32.0% live MAPE), highlighting model fragility.
- Directional accuracy remained inconsistent — arguably the most critical metric for practical trading decisions.

What I Would Improve:

- Incorporate additional features beyond closing prices: trading volume, technical indicators (RSI, MACD, Bollinger Bands), and sentiment analysis from news/social media.
- Ensemble methods that combine predictions from all three models for more robust forecasts.
- Walk-forward (rolling window) validation instead of a fixed train-test split.
- Market regime detection to adapt model parameters based on prevailing market conditions.
- Experiment with Temporal Fusion Transformers and larger datasets for the attention-based approach.